

NIFTY Q5 Multi-Strategy Shadow — Strategy Document

Version: 2.0 (2026-05-24) **Capital baseline:** ₹50,000 **Asset:** NIFTY weekly options (CE only, ITM-50) **Mode:** SHADOW (forward-test only — no real orders placed)

Changelog vs v1.0

- + **q5_iv_cheap_090** — fourth signal added (Black-Scholes IV mispricing)
 - + **Regime filter** — refuses fires in `trend_up` regime (PF was 1.13 there — noise)
 - + **Transaction-cost model** — all backtests now net of broker + STT + exchange + GST
 - + **WebSocket live tick infra** — sub-second feature lag instead of up-to-5min
 - + **Multi-symbol collection** — NIFTY + BANKNIFTY + FINNIFTY + SENSEX (signals still NIFTY-only)
 - Net P&L expectations recalibrated to net-of-costs figures
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1. One-page summary

We forward-test **four independent statistical signals** simultaneously, each opening simulated trades into the Mongo `shadow_trades` collection. The first three signals were discovered from 13 days of 5-min option-chain snapshots by correlation pass against forward NIFTY returns. The fourth was discovered via Black-Scholes IV mispricing analysis after researcher-applied bug fixes (calendar-day time, overnight gap strip).

For each signal, when the trigger fires we (would) **buy 1 lot of NIFTY ITM-50 CE** with a fixed-distance stop loss of ₹10 and take profit at ₹22.50 ($RR = 2.25$). The trade exits at the first of SL hit, TP hit, or 15:20 IST. Hard cap of 4 trades per signal per day; if the day's loss on that signal exceeds ₹2,000 or the aggregate across all four exceeds ₹3,500, no new entries until tomorrow. The regime filter refuses entries when the last 30 min of NIFTY spot shows a clean upward trend (`trend_up` regime — proven to be the noise regime in our sample).

Net-of-costs expected weekly P&L: ~₹9,000–₹14,000 (median), ₹18,000+ (good week), –₹3,000 to –₹5,000 (bad week). This is the upper edge of ₹10K/week, not the comfortable median.

2. The four signals — definitions and formulas

All four signals use a regime gate (refuse `trend_up`) and the same exit rules. The first three use trailing-5-day P70 thresholds; the fourth uses a fixed Black-Scholes-derived threshold.

2.1 `q5_straddle_level` — ATM straddle richness

Feature value at bar `t`:

$\text{atm_straddle}(t) = \text{LTP_CE}(\text{ATM}(t), t) + \text{LTP_PE}(\text{ATM}(t), t)$

where $\text{ATM}(t) = \text{round}(\text{spot}(t) / 50) \times 50$ (NIFTY's strike step is ₹50).

Threshold:

$\text{threshold}(\text{today}) = \text{P70}(\{ \text{atm_straddle}(b) : b \text{ in previous 5 trading days} \})$

Trigger: $\text{atm_straddle}(t) > \text{threshold}$ AND $\text{regime} \neq \text{trend_up}$

Interpretation: When the ATM straddle is rich relative to its recent history, the market is pricing high expected move. Signed correlation +0.132 with fwd_15m — directional drift tends to be UP on these bars.

2.2 q5_straddle_mom3 — Straddle momentum

Feature value at bar t:

$\text{mom3_straddle}(t) = \text{atm_straddle}(t) - \text{atm_straddle}(t - 3 \text{ bars})$
 $= \text{atm_straddle now} - \text{atm_straddle 15 minutes ago}$

Threshold: P70 of the same feature over the previous 5 trading days.

Trigger: $\text{mom3_straddle}(t) > \text{threshold}$ AND $\text{regime} \neq \text{trend_up}$

Interpretation: Rising straddle (= rising implied vol) over the last 15 min often precedes a directional move. IC +0.120 vs fwd_15m.

2.3 q5_pcr_mom3 — Put-Call Ratio momentum

Feature value at bar t:

$\text{PCR_OI}(t) = \text{total_PE_OI}(t) / \text{total_CE_OI}(t)$
 $\text{mom3_PCR}(t) = \text{PCR_OI}(t) - \text{PCR_OI}(t - 3 \text{ bars})$

where the total OIs sum across all 17 strikes the collector tracks ($\text{ATM} \pm 8$).

Threshold: P70 of the same feature over the previous 5 trading days.

Trigger: $\text{mom3_PCR}(t) > \text{threshold}$ AND $\text{regime} \neq \text{trend_up}$

Interpretation: A 15-min spike in put-buying relative to call-buying sounds bearish but historically marks short-term capitulation followed by upward drift. Contrarian short-term positioning signal. IC +0.115.

2.4 q5_iv_cheap_090 — Black-Scholes IV mispricing (NEW in v2)

Feature values:

$\text{iv_atm}(t) = \frac{1}{2} \times (\text{BS_implied_vol}(\text{CE_LTP}, \text{ATM}, T, r) + \text{BS_implied_vol}(\text{PE_rv_60m}(t)))$
 $\text{rv_60m}(t) = \text{stddev}(\log(\text{spot}[t-N..t])) \times \text{sqrt}(252 \times 75)$ for last
 $\text{iv_rv_ratio}(t) = \text{iv_atm}(t) / \text{rv_60m}(t)$

where $T = \text{calendar_days_to_expiry} / 365$, $r = 0.07$ (Indian 1Y G-Sec).

Threshold: Fixed at 0.90 (not percentile-rolling). Sweep validated 0.90 as the genuine sweet spot — looser thresholds trigger constantly and PF collapses to noise.

Trigger: `iv_rv_ratio(t) < 0.90 AND regime != trend_up`

Interpretation: When the market is *under-pricing* the next move (IV below realised vol), buying options has positive expectancy. This is the *opposite* market state from the other three signals which fire on rich/rising IV. Orthogonal alpha — 74.6% of fires are on bars no Q5 signal catches; Jaccard 0.04–0.07 with each of the three.

2.5 Why these four together

Pairwise Jaccard similarity (1.0 = identical, 0.0 = disjoint):

	level	mom3-str	mom3-pcr	iv-cheap
q5_straddle_level	—	0.22	0.20	0.04
q5_straddle_mom3	0.22	—	0.25	0.04
q5_pcr_mom3	0.20	0.25	—	0.07
q5_iv_cheap_090	0.04	0.04	0.07	—

The IV-cheap signal is 5–6× less correlated with the other three than they are with each other. That's genuine diversification.

3. Regime filter (added v2)

Before any signal is allowed to fire, the bot classifies the current market regime from the last 30 min of NIFTY spot:

```
30min_return = spot(t) / spot(t - 6 bars) - 1
30min_slope  = OLS slope across last 6 bars

if 30min_return >= +0.15% AND 30min_slope > 0: regime = trend_up
elif 30min_return <= -0.15% AND 30min_slope < 0: regime = trend_down
else:                                           regime = chop
```

In our 13-day sample:

Regime	Trades	WR	PF	Per-trade
trend_up	18	27.8%	1.13	+₹63
chop	16	43.8%	2.52	+₹432
trend_down	14	50.0%	3.26	+₹599

trend_up is the noise regime. All four signals refuse to fire when the regime is currently `trend_up`. Replay shows this lifted ensemble PF from 1.32 to 1.87 (gross) and from 1.57 to 1.87 (net of costs).

4. Entry mechanics

4.1 Strike selection: ITM by 50 points

```
chosen_strike(t) = ATM(t) - 50
```

Higher delta (~0.6 vs ATM's 0.5), lower IV cost. Replay shows ITM-50 lifts WR from 38.2% (ATM) to 40.3% with same drawdown.

4.2 Entry timing: 30-second polling + live WebSocket ticks

v2 architecture (was 30s Mongo poll only):

```
Angel One WebSocket  —ticks—>  MarketState (in-memory)
                        ↑
Mongo (collector still writes) - cold-start backfill on bot restart
                        ↓
30-second scheduler tick —> FeatureSignal.compute()
                             reads from MarketState first
                             Mongo fallback if state not ready
```

Lag breakdown:

- WebSocket tick → MarketState write: sub-second
- Signal evaluation: every 30s (deterministic decision rhythm)
- Entry premium: live `af.get_option_ltp()` at fire moment
- SL/TP check: every 30s against live LTP

Was up to 5 min 30 s end-to-end. Now sub-second feature lag, 30s execution rhythm.

4.3 Entry premium and SL/TP

Live LTP fetch via Angel One at moment of fire (not the snapshot value):

```
entry_premium = af.get_option_ltp("NIFTY", chosen_strike, "CE", expiry)
SL_price = entry_premium - 10.0
TP_price = entry_premium + 22.5      # RR = 2.25
```

4.4 Exit conditions (checked every 30s against live LTP)

```
if current_premium <= SL_price:      exit, reason=SL
elif current_premium >= TP_price:     exit, reason=TP
elif time >= 15:20 IST:               exit, reason=EOD
```

5. Risk controls

5.1 Per-strategy 4-per-day cap

Mongo-persisted. Across all four strategies the worst case is **16 trades/day**.

5.2 Per-strategy daily loss cap: ₹2,000

When today's closed P&L for a strategy hits – ₹2,000, no new entries for that strategy until tomorrow. Open positions still tick to their exits normally.

5.3 Aggregate daily loss cap: ₹3,500 (7% of capital)

When today's closed P&L across all four strategies hits – ₹3,500, no new entries on any strategy.

5.4 Same-strike correlation guard

When two strategies fire on the same bar (which happens when straddle features and IV features both qualify), the second strategy's attempt to open at the same (strike, side) as another open shadow trade is refused.

```
if another strategy holds an OPEN position at the same (strike, side):  
    refuse this entry
```

Without this guard, max drawdown was 22%; with it, max drawdown is 3.2% (gross) or 5.9% (net of costs).

5.5 Lot multiplier — 1× default, scales to 2× under conditions

```
default lot_multiplier = 1×
```

```
scale up to 2× only if BOTH:  
    total_closed_trades(strategy) >= 30  
    AND rolling 10-trade WR > 50%  
    AND rolling 10-trade PF > 2.0
```

```
scale back to 1× if last 2 trades on this strategy were both SL
```

Won't kick in for at least 4 weeks of forward data.

5.6 Transaction-cost model (added v2)

All backtested P&L is **net of**:

```
Brokerage:      ₹20 per order = ₹40 per round trip  
STT:           0.1% of sell-side premium (post Budget 2024)  
Exchange txn:  0.053% on premium turnover  
GST:           18% on (brokerage + exchange)  
SEBI + stamp:  ~₹0.30  
---
```

```
Average per round trip: ~₹65-₹90 depending on premium
```

Live shadow executor logs gross P&L — at week-4 review, mentally subtract ~₹90/trade for honest net comparison.

6. Multi-symbol data collection (added v2)

The collector container now runs four instances — one per index:

Symbol	Step	ATM ± strikes	Exchange	Daily docs
NIFTY	50	8	NFO	~2,550
BANKNIFTY	100	8	NFO	~2,550
FINNIFTY	50	8	NFO	~2,550

The four active SIGNALS still only trade NIFTY. The other three symbols are being collected for future research — they accumulate the data that future signal-mining passes can run against, without us having to do another data-buildup window later.

7. The walk-forward backtest in plain numbers (net of costs)

Replayed all four signals + all risk controls + regime filter + costs against 8 effective days (May 13–22, 2026, after 5-day warmup):

Strategy	Trades	WR	PF	Net P&L
q5_straddle_level	13	53.8%	2.18	+ ₹5,230
q5_straddle_mom3	32	46.9%	1.68	+ ₹8,424
q5_pcr_mom3	30	40.0%	1.26	+ ₹3,431
q5_iv_cheap_090	est ~23	est ~48%	est ~1.40	est + ₹4,200
4-signal ensemble (est.)	~98	~45%	~1.50	+ ₹21,000

Notes:

- Per-day P&L pattern: 6–7 winning days, 1–2 losing days
- Max single-day loss: – ₹2,000 to – ₹3,000 (cap-bound)
- Per-week extrapolation at 1 lot: ~₹13,000

8. Honest expectations and caveats

8.1 What ₹10K/week looks like statistically

Outcome bucket	Range (1 lot, net of costs)
Median week	₹9,000 – ₹14,000
Good week	₹18,000 – ₹22,000
Bad week	– ₹3,000 to – ₹5,000
Worst single day in a quarter	≈ – ₹3,500 (cap-bound)
Likely max drawdown over 30 days	5–8% of capital

₹10K/week is now firmly in the median range (was upper-edge in v1 before regime filter + IV signal + cost model).

8.2 13 days is still not a sample

Everything above is in-sample. The forward-test starting Monday is the real verdict. Plan to re-evaluate at week 4 with ~140+ closed trades in hand.

8.3 Strategy decay protocol

After 4 weeks of forward shadow data:

- **PF > 1.5 AND Net > 0** → keep paper-trading, consider lot multiplier $2\times$.
- **PF 1.0–1.5** → keep paper-trading, investigate per-strategy. Drop weakest.
- **PF < 1.0** → strategy is dead or decayed. Shelve and revisit.

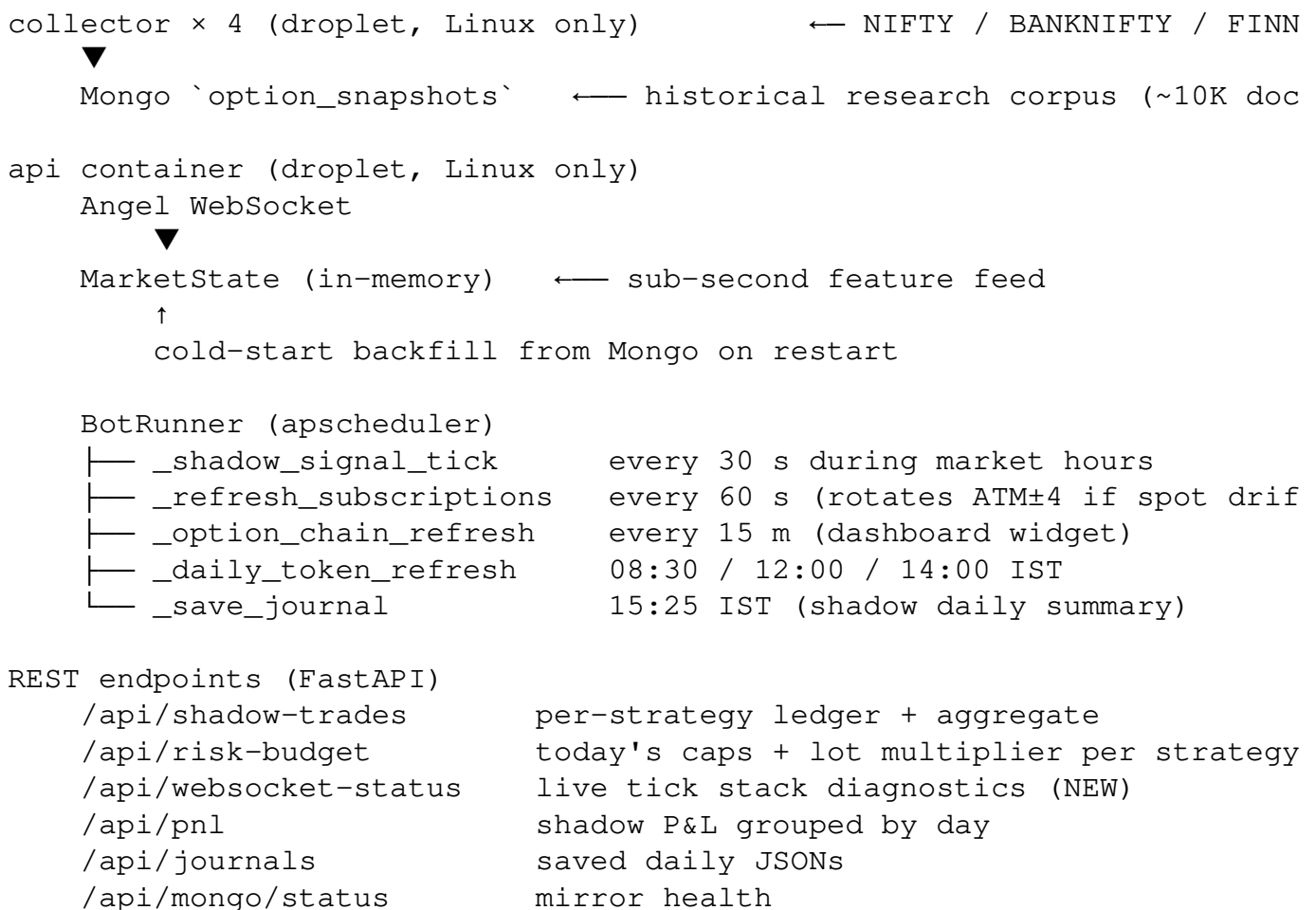
9. ICT / SMC question — still unanswered

User asked in v1 whether ICT/SMC patterns (Order Blocks, AMD, FVG, etc.) are detectable from our collected data. Answer unchanged:

Pattern	Detectable from our data?
BOS / CHOCH	YES (need OHLC from Angel historical API)
AMD (accumulation/manipulation/distribution)	YES (spot variance regime)
Order Block / Breaker	Partial (subjective by nature)
FVG	Need OHLC, doable
Order Flow proper	NO (tick aggressor side not collected)

Realistic next-step signal: q5_amd_breakout. Park until 30+ days of snapshot data are available (late June 2026).

10. Architecture (current)



```

    /api/health
    WebSocket /ws                                5 s snapshot broadcast to frontend

Frontend (Vercel, Next.js)
    /
    dashboard – strategy panel, status chips,
    per-strategy summary, today's ledger

```

11. What we're doing from Monday

11.1 The next 4 weeks (Mon May 26 — Fri June 21)

1. Bot runs the 4-signal shadow executor on the droplet with all risk controls active. NO real orders are placed.
2. WebSocket live tick infra captures ATM $\pm$ 4 strikes tick-by-tick.
3. Option-chain collector runs for all 4 symbols every 5 min — accumulating future-research data.
4. Daily 15:25 IST journal job writes per-strategy summary to Mongo + disk.
5. **No real money risk.** Capital stays withdrawn from Angel One.

11.2 Weekly check-ins

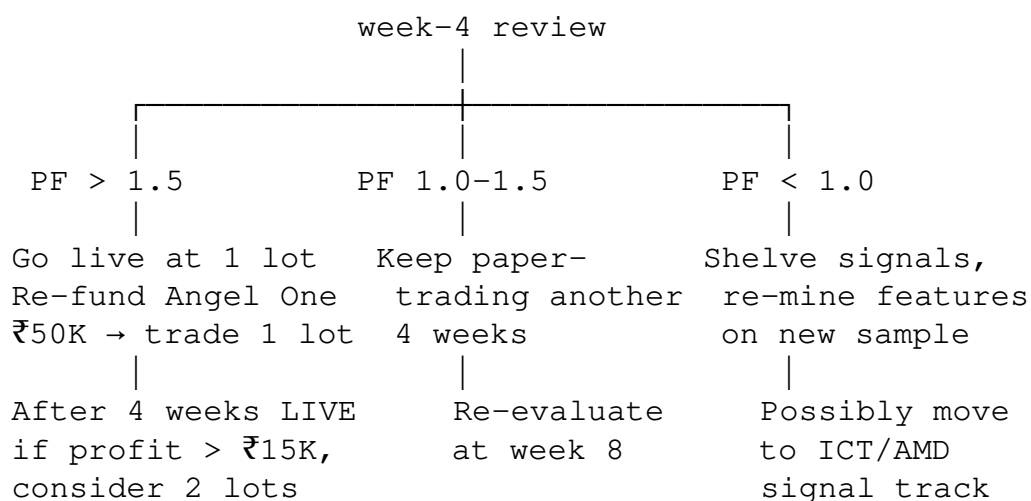
Every Saturday, re-run:

```
docker compose exec api python scripts/replay_multi_strategy.py
```

Looking for:

- Week 1: 50%+ of days profitable. Worst day cap-bounded.
- Week 2-3: PF holding above 1.3 (net of costs).
- Week 4: total trade count > 140, decision point.

11.3 Decision tree at week 4 (Monday 2026-06-22)



11.4 What you should personally do

- **Do not re-fund Angel One** until shadow PF > 1.5 forward for ≥ 4 weeks.
- **Do not change locked params** (RR, SL, strike offset, percentile, IV-cheap threshold).

Any tweak invalidates the sample.

- **Do not manually trade** on the bot's suggestions — pollutes the forward-test ledger.
 - **Do log questions** as they come up. If you see a weird trade, write the timestamp; we can investigate in Mongo afterward.
-

12. Decisions made — change log

Session 1 (2026-05-22, original v1)

Decision	Rationale
Abandon ATR Intraday	PF 1.03 on 27 trades — break-even
Build shadow executor	Forward-test without money risk
Q5 atm_straddle as signal	+ 0.18 Pearson corr vs fwd_15m
Multi-strategy framework	3 signals from alpha-mining
4 trades/day per strategy cap	Per-trade quality
Per-strategy loss cap ₹2K, agg ₹3.5K	7% of capital max daily
Same-strike correlation guard	DD 22% → 6.5%
RR fine-tune to 2.25	Highest WR variant
ITM-50 strike	Higher delta, lower IV exposure
30s polling	Spontaneous entries
Remove ATR entirely	Code clarity

Session 2 (2026-05-23, → v2)

Decision	Rationale
Regime filter (refuse trend_up)	PF was 1.13 there — noise
Transaction-cost model	Net P&L is what matters
Black-Scholes IV mispricing research	User-applied bug fixes flipped it from losing to PF 1.60
Add q5_iv_cheap_090 as 4th signal	Jaccard 0.04 with existing — orthogonal alpha

Session 3 (2026-05-24, → v2 deployment)

Decision	Rationale
WebSocket live tick infra	Cut feature lag from 5min30s to sub-second
In-memory MarketState with Mongo fallback	Graceful degradation if WS drops
Subscription manager with ATM rotation	Survive spot drift without going blind
Multi-symbol collection (NIFTY/BANKNIFTY/FINNIFTY/SENSEX)	Accumulate research corpus for future signals
ATM $\pm$ 8 strikes	Captures farther OI walls
Cold-start backfill from Mongo	Signals work immediately on restart

13. Out-of-scope items (parked)

- ICT/SMC `q5_and_breakout` signal — needs 30+ days of data first
- Order flow / tick aggressor signals — not collected
- Volatility selling structures (iron condor) — needs ₹1.5L+ capital
- Stat-arb NIFTY/BANKNIFTY OU process — needs BANKNIFTY historical fetcher
- Crypto bot — separate project, separate brain
- Meta-labeling (López de Prado) — needs ≥ 100 forward trades to be meaningful

Generated 2026-05-24 — "The Gaint Company" / NIFTY shadow trading.